

CosineSimilarity#

<https://pytorch.org/docs/stable/generated/torch.nn.modules.distance.CosineSi>

Description:

Returns cosine similarity between x_1 and x_2 , computed along dim . dim (int, optional) – Dimension where cosine similarity is computed. Default: 1 eps (float, optional) – Small value to avoid division by zero. Default: $1e-8$ Input1: $(\ast 1, D, \ast 2)$ where D is at position dim Input2: $(\ast 1, D, \ast 2)$, same number of dimensions as x_1 , matching x_1 size at dimension dim , and broadcastable with x_1 at other dimensions.

Function Signature

```
class torch.nn.modules.distance.CosineSimilarity(dim=1, eps=1e-08)[source]#
```

Parameters

Shape:

```
forward(x1, x2)[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> input1 = torch.randn(100, 128)
>>> input2 = torch.randn(100, 128)
>>> cos = nn.CosineSimilarity(dim=1, eps=1e-6)
>>> output = cos(input1, input2)
```

Detailed Documentation

Documentation

Returns cosine similarity between x_1x_1 and x_2x_2 , computed along dim.

dim (int, optional) – Dimension where cosine similarity is computed. Default: 1

eps (float, optional) – Small value to avoid division by zero. Default: 1e-8

Input1: $(*1,D,*2)(\backslash ast_1, D, \backslash ast_2)(*1,D,*2)$ where D is at position dim

Input2: $(*1,D,*2)(\backslash ast_1, D, \backslash ast_2)(*1,D,*2)$, same number of dimensions as x_1 , matching x_1 size at dimension dim, and broadcastable with x_1 at other dimensions.

Output: $(*1,*2)(\backslash ast_1, \backslash ast_2)(*1,*2)$

Runs the forward pass.